

Sushma Reddy Vutla

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Data Analyst Intern | Spesh Talent | TX

Aug 2024 - Dec 2024

- Gathered, cleaned, and validated datasets from multiple client and internal systems using **SQL** and **Python (Pandas)** to ensure accurate reporting for consulting projects.
- Performed **exploratory analysis** and developed dashboards in **Tableau** to visualize KPIs, enabling faster decision-making.
- Built and optimized **SQL queries** to support marketing and operational reporting, improving query performance and reducing dashboard refresh times by **20%**.
- Partnered with cross-functional teams to translate analytical results into **actionable insights** aligned with client business objectives.

Software Engineer - Client: AXA UK | Cognizant | India

Mar 2021 - Jun 2023

Project: AXA UK - BAU Run Application Management & Development

- Worked across the full Software Development Lifecycle (SDLC)** from requirements gathering, design, coding, testing, and documentation to deployment and production support ensuring delivery of solutions that met **strict internal controls and regulatory requirements**.
- Developed, enhanced, and deployed **RESTful APIs and Java/Spring Boot microservices** (Maven, Git) integrating with front-end and third-party systems; delivered end-to-end application management and production support for 12+ services, achieving **99.9% uptime** through SLA adherence, proactive monitoring, patching, and incident triaging in ServiceNow.
- Designed and executed **robust testing suites** including **unit testing (JUnit)**, **integration testing (SoapUI)**, and **regression testing** and implemented User Acceptance Testing (**UAT**) and **production deployments** by creating change tickets, performing impact analysis, scheduling releases, and executing validations ensuring zero critical post-release defects and 100% release success.
- Executed **BAU operations** including batch monitoring, middleware maintenance, outage resolution, license management, and database operations; implemented **robust logging and exception handling** (SLF4J, Log4j), accelerating root cause analysis and reducing production debugging time by **40%**.
- Optimized **SQL queries and compliance-driven reports** (Oracle, Active Directory) for monthly, quarterly, and ad-hoc business needs, improving performance by **20%** while maintaining 100% accuracy.
- Strengthened **security and compliance posture** by upgrading JARs, resolving dependency issues, and performing quarterly patching, DR drills, and Pentest remediation, reducing vulnerabilities by **30%** and ensuring audit readiness.
- Authored **SOPs, deployment guides, and Knowledge Base (KEDB) documentation**, reducing onboarding time by **30%**, and collaborated with **cross-functional teams (QA, Infrastructure, UAT, Vendors)** under Agile methodology to refine user stories, streamline release cycles, and deliver enhancements with zero critical post-release defects.

Data Analyst | HRH NEXT Services | India

May 2020 - Feb 2021

- Consolidated operational and customer service data from multiple systems using **SQL** to enable accurate trend analysis and unified reporting for call center performance.
- Designed and developed Power BI dashboards to monitor SLA adherence, call volumes, and agent performance, providing **real-time operational visibility** for managers.
- Conducted trend and variance analysis using SQL and **Excel** to identify drivers of efficiency, enabling workflow improvements.
- Performed **regression analysis** on call trends using **Python**, reducing escalations by **12%** through data-informed process changes.
- Delivered **clear, client-ready reports** by translating analytical outputs from **Power BI** into actionable recommendations.

Data Engineer Intern | PAteam | India

Dec 2019 - Apr 2020

- Assisted in developing ETL workflows using **SQL Server** and Python (Pandas) to extract and prepare **CRM** and operational data for reporting.
- Supported **SQL query improvements** under guidance of senior engineers, improving dashboard data retrieval times.
- Documented **data processes** and ran Python-based **data quality checks** to verify dataset accuracy for analytics use cases.

EDUCATION

- The University of Texas at Dallas - Naveen Jindal School of Management** Aug 2023 - May 2025
Master of Science in Business Analytics and Artificial Intelligence GPA: 3.72
- Jawaharlal Nehru Technological University, Hyderabad** Aug 2016 - May 2020
Bachelor of Technology in Mechanical Engineering GPA: 3.4

SKILLS

Languages	Core Java, Python, R, SQL
Frameworks & Libraries	Spring Boot, Spring Framework, REST/SOAP APIs, Web Services, SLF4J, Log4j, JUnit, ggplot2, Matplotlib, Seaborn, tidyverse, caret, rpart, neuralnet
Databases	SQL, Oracle DB, MySQL, SQL Server, Hadoop, Spark, Hive, Impala
Cloud Platforms (AWS)	S3, Glue, Lambda, Athena, Quick Sight, Cloud Computing Concepts
Concepts & Pipelines	ETL Pipelines, Data Warehousing, Data Modeling, Data Lakes, ER Modeling, Star Schema, OLAP/OLTP
Machine Learning	Supervised & Unsupervised Learning, Predictive Modeling, Prescriptive Analytics, Decision Trees, Logistic Regression, Neural Networks, Model Validation, Cross-Validation, Bias-Variance Tradeoff
Statistical Analysis	Econometrics (OLS, 2SLS, IV Regression, Hausman Test), A/B Testing & RCTs (ROAS, CTR, CAC), Exploratory Data Analysis (EDA), Hypothesis Testing, ANOVA, Regression Analysis
Visualization	Tableau, ggplot2, Matplotlib, Seaborn
Tools	ServiceNow (Incident/Problem/Change Management), Silva, Maven, Putty, WinSCP, Nexus Repository
Methodologies	Agile Methodology, ITIL Processes, Cross-Functional Stakeholder Management (QA, UAT, Infra, Vendors), Release Management, Deployment Validation, Project Collaboration
Security & Compliance	Vulnerability Remediation, SSL Certificate Management, Dependency Upgrades

ACADEMIC PROJECTS

- **Real-Time Data Lakehouse & ETL Pipeline on AWS** | AWS S3, Glue, Lambda, Athena, QuickSight, SQL
*Designed and implemented a **scalable ETL pipeline in AWS** to automate processing of YouTube trending data; used AWS Glue for cataloging & transformation, optimized queries with Athena, and delivered **dashboards** for **real-time analytics**.*
 - **Big Data Risk Analytics with Hadoop & Spark** | Hadoop, Spark, Hive, Impala, Tableau, JDBC
*Analyzed **7+ real-world fleet datasets** to identify high-risk drivers, violation hotspots, and insurance risks for AZ National Trucking; built and automated **15+ live Tableau dashboards** connected to a Hadoop cluster via JDBC.*
 - **Public Health Data Analytics with Python & Tableau** | Python, SQL, Tableau, Data Cleaning, EDA, Data Visualization
*Cleaned and analyzed **CDC county-level COVID-19 data** to identify geographic, demographic, and socioeconomic factors influencing vaccine hesitancy; automated workflows in Python and built **interactive Tableau dashboards** to support **targeted public health recommendations**.*
 - **Machine Learning Models for Socioeconomic Profiling** | R, tidyverse, caret, rpart, neuralnet, ggplot2
*Built **Decision Tree, Logistic Regression, and Neural Network models** on UCI Adult Census data to predict income levels (>50K vs ≤50K), achieving **~81% accuracy** and generating interpretable insights to inform **policy and resource allocation**.*
 - **Econometric Analysis of Income Inequality** | R, Econometrics (OLS, 2SLS, IV Regression), Hausman Test, Inequality Analysis
*Applied **advanced econometric techniques** (OLS, 2SLS, IV Regression) in R to estimate the causal impact of education and occupation on income; addressed endogeneity with instrumental variables and subgroup analysis, producing **robust insights on income inequality**.*
 - **Relational Database & BI for Delivery Optimization** | SQL (MySQL), Tableau, Data Modeling, Business Intelligence (BI)
*Designed a **normalized MySQL database** for a courier delivery service, developed advanced **SQL queries** to track KPIs, and created **interactive Tableau dashboards** to optimize **delivery times, courier allocation, and customer satisfaction**.*
- 🔴 **Practical Experience - Causal Analytics & A/B Testing**
*Designed and executed **A/B tests** and **RCTs** for marketing campaigns (Vungle Ads, Uber Express POOL, BIT, Behavior Change for Good) to evaluate **ROAS, CTR, CAC** and measure treatment effects.*